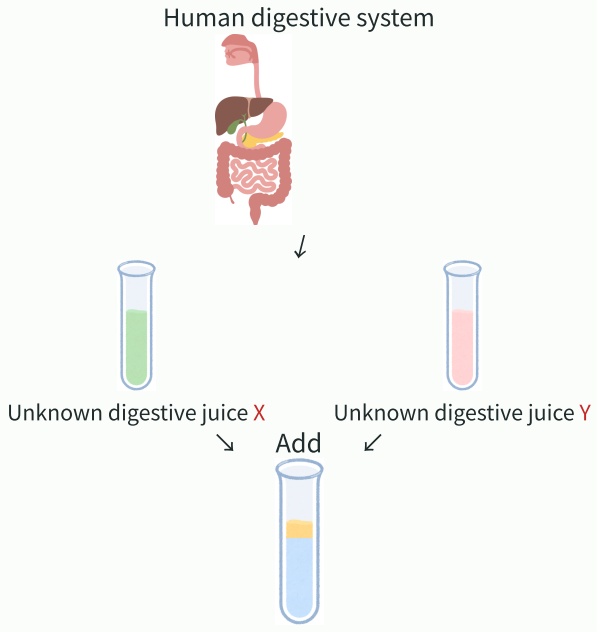
Name: _____ Class and student number: _____

Digestion of lipids and fair comparisons

Lipids in food must be digested into small molecules before absorption.
Four sets compare adding X, Y or both to an oil–water mixture.
The four setups receive equal total added volumes and use the same conditions.
Use the notes and figures to complete B, then check with C. Keep A available.



Main experiment setup (before observation)

Keep starting oil/water amounts, total added volume (2 mL), 37°C, observation time and shaking the same. Added liquids are shown in the figure.

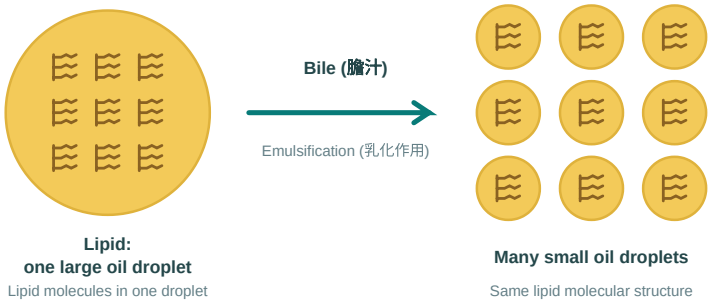
Setup 1	Setup 2	Setup 3	Setup 4
Water set: add 2 mL water	Add 2 mL of digestive juice X	Add 2 mL of digestive juice Y	Add 1 mL each of digestive juices X and Y



Supporting notes: physical and chemical digestion

01 • Physical digestion (物理消化)

Bile (膽汁), made in the liver and stored in the gall bladder (膽囊), has no digestive enzymes. Emulsification (乳化作用) makes smaller droplets with more total surface area; lipid molecules stay unchanged.

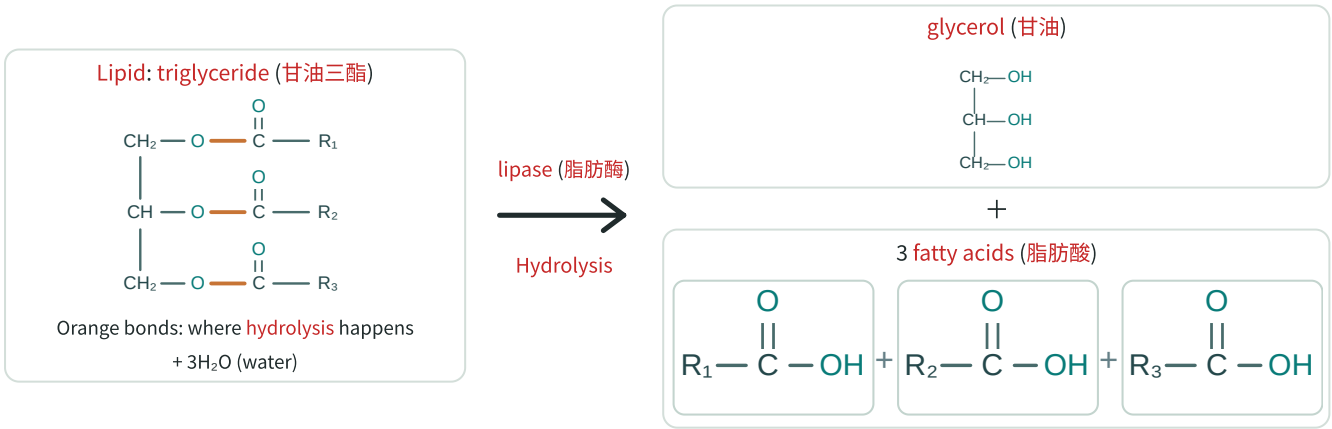


Lipids (large oil droplets) → Lipids (small oil droplets)

Lipid molecules stay the same during emulsification (乳化作用). Not to scale.

02 • Chemical digestion (化學消化)

Lipase (脂肪酶) catalyses hydrolysis, changing chemical bonds to form glycerol (甘油) and fatty acids (脂肪酸). The enzyme is not used up. This course uses a simplified complete-hydrolysis model.



Triglyceride (甘油三酯) + $3H_2O$ (water) → glycerol (甘油) + 3 fatty acids (脂肪酸)

Complete-hydrolysis model. R_1–R_3 are fatty acid (脂肪酸) chains; orange bonds show where hydrolysis occurs.